

CAPSTONE PROJECT & TEST PLAN

Project Name: E-Commerce Platform Automation Framework

Automation Tool: Playwright (JavaScript/TypeScript)

Target Website: LambdaTest Ecommerce Playground (<https://ecommerce-playground.lambdatest.io/>)

Total Services to be Tested: 8

Test Cases Count: 15 Tests per service (Total: 120 Test Cases)

PROJECT OBJECTIVE & OVERVIEW

The objective of this Capstone project is to build a robust, end-to-end automation testing framework using Playwright. The framework will validate 8 core services of the e-commerce platform across different functional layers. A total of 120 unique test cases (15 per service) will be designed and executed to ensure the stability, reliability, and performance of the application.

SELECTED SERVICES FOR AUTOMATION

1. Authentication & User Management

- **Description:** This service handles user access control, validation, and session maintenance.
- **Scope of Testing:** Verifying user registration, successful login with valid credentials, login failures with invalid inputs, "Forgot Password" redirection, blank field validations, and explicit user logout scenarios.

2. Product Search & Discovery

- **Description:** This service allows users to look up and filter items across the e-commerce store.
- **Scope of Testing:** Validating global search bar accuracy, partial text search matching, filtering mechanisms (by price range and brand categories), and sorting functionalities (such as Low to High, High to Low, and Latest arrivals).

3. Cart Management

- **Description:** This service manages the user's shopping bag, computing totals, and tracking item quantities.
- **Scope of Testing:** Verifying "Add to Cart" operations, real-time cart badge count increments, modifying quantities within the cart, deleting items, subtotal recalculations, and coupon code application logic.

4. User Profile Management

- **Description:** This service handles individual user account settings, historical records, and communication preferences.
- **Scope of Testing:** Validating personal detail updates (Name, Email), changing account passwords, verifying password validation criteria, and checking order history dashboards.

5. Address Book & Shipping Service

- **Description:** This service manages the delivery logistics and multi-address capabilities for a user.
- **Scope of Testing:** Verifying the addition of new shipping/billing addresses, input validations for mandatory fields (such as Zip Codes and Phone numbers), updating or deleting existing addresses, and configuring a default shipping address.

6. Customer Support & Feedback

- **Description:** This service covers the user communication channels and customer help interfaces.
- **Scope of Testing:** Testing form submissions on the "Contact Us" page, validating mandatory field restrictions on support forms, and testing user interactions with the Frequently Asked Questions (FAQ) expand/collapse accordions.

7. Internal API Testing

- **Description:** This service validates the application's underlying backend communication layers directly using Playwright's API capabilities.
- **Scope of Testing:** Verifying HTTP status codes (like 200 OK or 401 Unauthorized) for core endpoints, asserting backend JSON payload structures, and testing error payloads against invalid data structures.

8. System Persistence & State Validation

- **Description:** This service ensures that application data remains consistent across user sessions and environmental interruptions.
- **Scope of Testing:** Verifying that cart items and login sessions persist after explicit page reloads, testing application behavior after hard refreshes, and validating storage states (Cookies and Local Storage).